EXERCISES COURSE

LING WANG

Only For Teaching! Continues Updating!
Personal website: https://lwmath.github.io

1. First class (9.19)

Problem 1.1 (1.3)**.** *For* $\forall\, n \in \mathbb{N}$*, let* $A_n = \left[-1 + \frac{1}{2n}, 1 - \frac{1}{n}\right]$*, prove* $\bigcup\limits_{n=1}^{\infty} A_n = (-1, 1)$*.*

Proof. Firstly, by definition we have that

$$\forall\, x \in \bigcup_{n=1}^{\infty} A_n, \quad \exists\, n_0 \in \mathbb{N}, \quad s.t.\ x \in A_{n_0} \subset (-1, 1).$$

Hence

$$x \in (-1, 1),$$

i.e.

$$\bigcup_{n=1}^{\infty} A_n \subset (-1, 1). \tag{1.1}$$

On the other hand, $\forall\, x \in (-1, 1)$, set

$$N = \left[\max\left\{\frac{1}{2(x+1)}, \frac{1}{1-x}\right\}\right] + 1,$$

then there is $x \in A_N$, thus $x \in \bigcup\limits_{n=1}^{\infty} A_n$, which means

$$(-1, 1) \subset \bigcup_{n=1}^{\infty} A_n. \tag{1.2}$$

Combining (1.1) and (1.2) yields

$$\bigcup_{n=1}^{\infty} A_n = (-1, 1).$$

□

Problem 1.2 (1.5)**.** *Prove* $A = \left\{n \sin \frac{n\pi}{2}; n \in \mathbb{Z}\right\}$ *is unbounded.*

Date: September 24, 2022.

Proof. **Definition:** A set X is unbounded if and only if $\forall\, M > 0$, there exists a $x \in X$, such that $|x| > M$. Then $\forall\, M > 0$, let

$$N = 2([M] + 1) + 1,$$

hence

$$\left| N \sin \frac{N\pi}{2} \right| = 2([M] + 1) + 1 > M,$$

which implies that A is unbounded. □

Problem 1.3 (1.23)**.** *Suppose $f(x)$ is defined on E, where $|E| \geq 3$. Prove $f(x)$ is strictly monotonic on E iff for $\forall\, x_1, x_2, x_3 \in E$, if $x_1 < x_2 < x_3$, then there must be*

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) > 0. \tag{1.3}$$

Proof. " only if ": Obviously.

" if ": Suppose that $f(x)$ is not a strictly monotonic function, hence there exist x_1, x_2, x_3, $x_4 \in E$ with $x_1 < x_2$ and $x_3 < x_4$ such that

$$f(x_1) \leq f(x_2) \quad \text{and} \quad f(x_3) \geq f(x_4).$$

If $x_1 < x_2 < x_3 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) \quad \text{and} \quad (f(x_2) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

If $x_1 < x_3 < x_2 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_4)) \quad \text{and} \quad (f(x_1) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

Similar for other cases. Then we obtain contradictions for all cases, wihch means that $f(x)$ is strictly monotonic on E. □

Problem 1.4 (1.25)**.** *Prove $\sin(x^2 + x)$ is not a periodic function.*

Proof. Suppose $T > 0$ is the period of $f(x) := \sin(x^2 + x)$, we have that

$$f(T) = f(-T) = f(0) = 0.$$

Hence there are $k_1, k_2 \in \mathbb{Z}$ such that

$$T^2 + T = k_1\pi \quad \text{and} \quad T^2 - T = k_2\pi,$$

which yields

$$T = \frac{k_1 - k_2}{2}\pi.$$

Then

$$\frac{(k_1 - k_2)^2}{4}\pi + \frac{k_1 - k_2}{2} = k_1,$$

i.e.

$$\pi = \frac{2(k_1 + k_2)}{(k_1 - k_2)^2} \in \mathbb{Q},$$

contradiction. □

Problem 1.5 (1.26)**.** *Suppose $f(x)$ is defined on $(0, +\infty)$, $x_1, x_2 > 0$. Prove*

(1) *If $\frac{f(x)}{x}$ is decreasing, then $f(x_1 + x_2) \leq f(x_1) + f(x_2)$;*
(2) *If $\frac{f(x)}{x}$ is increasing, then $f(x_1 + x_2) \geq f(x_1) + f(x_2)$.*

Proof. (1) Since $\frac{f(x)}{x}$ is decreasing, then $f(x_1 + x_2) \leq f(x_1) + f(x_2)$, we have

$$\frac{f(x_1 + x_2)}{x_1 + x_2} \leq \frac{f(x_1)}{x_1} \quad \text{and} \quad \frac{f(x_1 + x_2)}{x_1 + x_2} \leq \frac{f(x_2)}{x_2}.$$

Hence

$$f(x_1 + x_2) = \frac{x_1}{x_1 + x_2} f(x_1 + x_2) + \frac{x_2}{x_1 + x_2} f(x_1 + x_2) \leq f(x_1) + f(x_2).$$

Similar for (2). □

Problem 1.6 (1.27)**.** *Suppose $f(x)$ is defined on $(-\infty, +\infty)$, and $f(f(x)) \equiv x$.*

(1) *Is $f(x)$ unique? If not, please give an example;*
(2) *If $f(x)$ is strictly increasing, is it unique? Why?*

Solution. (1) $f(x)$ is not unique. For example $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}, x \neq 0$; $f_3(0) = 0$.

(2) We prove that $f(x) = x$. Suppose not, if there is a $x_0 \in \mathbb{R}$, such that $f(x_0) \neq x_0$. Without loss of generality, we assume that $f(x_0) > x_0$. Then $f(f(x_0)) > f(x_0)$ by f is strictly increasing. Hence

$$x_0 = f(f(x_0)) > f(x_0) > x_0,$$

contradiction. □

Problem 1.7. *Prove that $f(x) = \sin x + \sin \sqrt{2}x$, $x \in \mathbb{R}$ is not a periodic function.*

Proof. Note that

$$f''(x) = -\sin x - 2 \sin \sqrt{2}x.$$

Hence we have

$$\begin{aligned} f(x) + f''(x) &= -\sin \sqrt{2}x, \\ 2f(x) + f''(x) &= \sin x. \end{aligned}$$

If $T > 0$ is the period of $f(x)$, then T is also the period of $f(x) + f''(x)$ and $2f(x) + f''(x)$. Hence

$$T = \frac{2k\pi}{\sqrt{2}} = 2m\pi, \quad \text{for some } k, m \in \mathbb{N},$$

which implies

$$\sqrt{2} = \frac{k}{m} \in \mathbb{Q},$$

contradiction. □

Problem 1.8. *Suppose $f(x)$ is an increasing function defined on closed interval $[a, b]$, and f satisfies $f(a) \geq a$, $f(b) \leq b$. Prove that there exists $x_0 \in [a, b]$, such that $f(x_0) = x_0$.*

Proof. Set

$$A = \{x \in [a, b] : f(x) \geq x\}.$$

Obviously, A is not empty since $f(a) \geq a$, and b is an upper bound of A. Hence $\sup A$ exists. We denote $x_0 := \sup A$. For $\forall\, x \in A$, there is $x \leq x_0$. Since f is increasing, we have $x \leq f(x) \leq f(x_0)$, i.e. $f(x_0)$ is an upper bound of A, thus $x_0 \leq f(x_0)$. If $x_0 < f(x_0)$, then $f(x_0) \leq f(f(x_0))$, which means $f(x_0) \in A$, contradiction. Hence $f(x_0) = x_0$. □

Problem 1.9. *Is there a function, whose period are all rational numbers but none of irrational number?*

Solution. Dirichlet function, i.e.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \backslash \mathbb{Q}. \end{cases}$$

□

2. Second class (9.26)

Problem 2.1 (2.3(5))**.** *Using definition to prove*

$$\lim_{n \to \infty} n^3 q^n = 0 \, (|q| < 1).$$

Proof. It suffices to prove $\lim\limits_{n \to \infty} n^3 |q|^n = 0$, and it is obviously valid when $|q| = 0$, hence we assume that $|q| \neq 0$ in the following. Let

$$\frac{1}{|q|} = 1 + \alpha.$$

By binomial theorem, we have

$$\frac{1}{|q|^n} = (1 + \alpha)^n = \sum_{k=0}^{n} C_n^k \alpha^k \geq \frac{n(n-1)(n-2)(n-3)}{4!} \alpha^4$$

provided with $n \geq 4$. Hence for $n \geq 4$,

$$n^3 |q|^n \leq \frac{24 n^2}{(n-1)(n-2)(n-3)\alpha^4} < \frac{72}{\alpha^4} \frac{1}{n-3}.$$

Then $\forall\, \varepsilon > 0$, choosing $N = \left[\frac{72}{\alpha^4 \varepsilon}\right] + 4$, for any $n > N$, there is $n^3 |q|^n < \varepsilon$, i.e. $\lim\limits_{n \to \infty} n^3 |q|^n = 0$. □

Problem 2.2 (2.4)**.** *Suppose for all $n \in \mathbb{N}$, there is $x_n \leq a \leq y_n$ and $\lim\limits_{n \to \infty} (y_n - x_n) = 0$. Prove*

$$\lim_{n \to \infty} x_n = a = \lim_{n \to \infty} y_n.$$

Proof. Since $x_n \le a \le y_n$, we know that

$$0 \le a - x_n \le y_n - x_n.$$

By Sandwich Theorem, we have

$$\lim_{n\to\infty} x_n = a.$$

Similar for y_n. □

Problem 2.3. *Suppose $\{a_n\}$ is monotonic increasing, $\{b_n\}$ is monotonic decreasing, and $\lim_{n\to\infty}(b_n - a_n) = 0$. Prove that $\lim_{n\to\infty} a_n$, $\lim_{n\to\infty} b_n$ exist, and they are equal.*

Proof. Assume $\lim_{n\to\infty} a_n$ doesn't exist, since $\{a_n\}$ is monotonic increasing, we know there is $\lim_{n\to\infty} a_n = +\infty$. By $\{b_n\}$ is monotonic decreasing, and $\lim_{n\to\infty}(b_n - a_n) = 0$, there must be $\lim_{n\to\infty} b_n = -\infty$. Hence, for $\forall\, M > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N$, there are

$$a_n > M \quad \text{and} \quad b_n < -M.$$

Then

$$b_n - a_n < -2M,$$

which contradicts with $\lim_{n\to\infty}(b_n - a_n) = 0$. Thus, $\lim_{n\to\infty} a_n$ exists. It's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. □

Problem 2.4 (2.6(3))**.** *$\{F_n\}$ is the Fibonacci sequence, defined by*

$$F_0 = F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}.$$

Prove $\lim_{n\to\infty} F_n = +\infty$ by definition.

Proof. **First way:** We can calculate the general terms of F_n by 'eigenvalue method' (See Problem 2.10 for details). Then we know

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right].$$

Since $\frac{1+\sqrt{5}}{2} > 1$ and $\frac{1-\sqrt{5}}{2} < 1$, we have $\lim_{n\to\infty} F_n = +\infty$.

Second way: We can prove by induction that $F_n \ge n$ for $n \ge 1$. Since $\lim_{n\to\infty} n = +\infty$, we know $\lim_{n\to\infty} F_n = +\infty$. □

Problem 2.5 (2.10(6))**.** *Calculate*

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}).$$

Solution.

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}) = \lim_{n\to\infty} \frac{\sqrt[3]{n}}{(\sqrt[3]{n+1})^2 + \sqrt[3]{n+1}\sqrt[3]{n} + (\sqrt[3]{n})^2} = 0.$$

□

Problem 2.6 (2.14)**.** *Calculate* $\lim_{n\to\infty} x_n$.

(1) $x_n = \dfrac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$;

(2) $x_n = \displaystyle\sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}}$;

(3) $x_n = \sqrt[n]{n\ln n}$.

Solution. (1)Note that

$$\frac{q}{p} < \frac{q+1}{p+1}, \quad \text{for } 0 < q < p.$$

Then

$$\begin{aligned} x_n &= \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)} \\ &= \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n} \\ &< \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{2n-2}{2n-1}\cdot\frac{2n}{2n+1} \\ &= \frac{2}{1}\cdot\frac{4}{3}\cdot\frac{6}{5}\cdots\frac{2n}{2n-1}\cdot\frac{1}{2n+1} \\ &= \frac{1}{x_n}\frac{1}{2n+1}, \end{aligned}$$

which gives us

$$x_n < \frac{1}{\sqrt{2n+1}}.$$

Hence $\lim_{n\to\infty} x_n = 0$.

(2) Note that $(n+1)^2 - n^2 + 1 = 2(n+1)$, we have

$$2 = \frac{2(n+1)}{n+1} \le \sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}} \le \frac{2(n+1)}{n} \to 2, \quad \text{as } n\to\infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 2$.

(3) When $n > 3$, there is $n < n\ln n < n^2$. Then

$$1 \leftarrow \sqrt[n]{n} < \sqrt[n]{n\ln n} < \sqrt[n]{n^2} \to 1, \quad \text{as } n\to\infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 1$. $\square$

Problem 2.7 (2.17)**.** *Sequence* $\{q_n\}$ *satisfies*

$$0 < q_n < 1, \quad (1-q_n)q_{n+1} > \frac{1}{4}, \quad \forall\, n \in \mathbb{N}. \tag{2.1}$$

Prove that $\{q_n\}$ *is monotonic increasing and* $\lim\limits_{n\to\infty} q_n = \frac{1}{2}$.

Proof. By (2.1) and the mean value inequality, we have

$$\frac{q_{n+1}}{q_n} > \frac{1}{4q_n(1-q_n)} \geq \frac{1}{4\left(\frac{q_n+1-q_n}{2}\right)^2} = 1,$$

i.e. $q_{n+1} > q_n$. Hence $\{q_n\}$ is monotonic increasing. Then by the monotone bounded convergence theorem, we know $\lim\limits_{n\to\infty} q_n$ exists. Denote $q := \lim\limits_{n\to\infty} q_n$, by (2.1),

$$(1-q)q \geq \frac{1}{4},$$

i.e.

$$\left(q - \frac{1}{2}\right)^2 \leq 0,$$

which means $q = \frac{1}{2}$. □

Problem 2.8 (2.19)**.** *Suppose that* $0 < a_1 < b_1$, *let*

$$a_{n+1} = \sqrt{a_n b_n}, \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \quad (n = 1, 2, \cdots).$$

Prove $\lim\limits_{n\to\infty} a_n$ *and* $\lim\limits_{n\to\infty} b_n$ *exist, and* $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n$.

Proof. By the mean value inequality, we have

$$a_{n+1} = \sqrt{a_n b_n} \leq \frac{1}{2}(a_n + b_n) = b_{n+1} \quad \text{for } n \geq 1.$$

Hence

$$a_{n+1} = \sqrt{a_n b_n} \geq a_n \quad \text{and} \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \leq b_n \quad \text{for } n \geq 2.$$

Then

$$a_2 \leq \cdots \leq a_n \leq b_n \leq \cdots \leq b_2 \quad \text{for all } n \geq 2.$$

By the monotone bounded convergence theorem, we know $\lim\limits_{n\to\infty} a_n$ and $\lim\limits_{n\to\infty} b_n$ exist, and it's easy to see that $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n$. □

Problem 2.9. *Suppose that* $a_1 > b_1 > 0$, *let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{2a_n b_n}{a_n + b_n} \quad (n = 1, 2, \cdots).$$

Prove $\lim\limits_{n\to\infty} a_n$ *and* $\lim\limits_{n\to\infty} b_n$ *exist, and* $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \sqrt{a_1 b_1}$.

Proof. The existence is similar to Problem 2.8. Thus, we only need to prove $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$, and if we notice that

$$a_{n+1}b_{n+1} = a_n b_n = \cdots = a_1 b_1,$$

and

$$a_{n+1} \geq \sqrt{a_n b_n} \geq b_{n+1},$$

it's easy to obtain the conclusion. □

Problem 2.10. *Suppose that $a_1 = \alpha$, $b_1 = \beta$. Let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{a_{n+1} + b_n}{2} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$, and find the limitation.

Proof. By $a_{n+1} = (a_n + b_n)/2$, we have $b_n = 2a_{n+1} - a_n$. Hence

$$2a_{n+2} - a_{n+1} = \frac{a_{n+1} + 2a_{n+1} - a_n}{2} = \frac{3}{2}a_{n+1} - \frac{1}{2}a_n,$$

which is

$$a_{n+2} = \frac{5}{4}a_{n+1} - \frac{1}{4}a_n. \tag{2.2}$$

Then the characteristic equation of (2.2) is

$$x^2 - \frac{5}{4}x + \frac{1}{4} = 0.$$

We find that $x = 1$ and $x = \frac{1}{4}$ are the solution. Hence the general form of a_n is

$$a_n = A + B\left(\frac{1}{4}\right)^n.$$

By $a_1 = \alpha$, $a_2 = (a_1 + b_1)/2 = (\alpha + \beta)/2$, we have

$$a_n = \frac{1}{3}\alpha + \frac{2}{3}\beta + \frac{2}{3}(\alpha - \beta)\left(\frac{1}{4}\right)^{n-1}.$$

Hence $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \frac{1}{3}\alpha + \frac{2}{3}\beta$. □

Problem 2.11 (2.20)**.** *Calculate following limitations.*

(2) $\lim_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n$;

(3) $\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2}$.

Solution. (2)We first note that

$$\left(1+\frac{1}{n^2}\right)^{n^2} < e.$$

(This can be proved by definition of e.) Then we have

$$1 < \left(1+\frac{1}{n^2}\right)^{n} < e^{\frac{1}{n}} \to 1 \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem

$$\lim_{n\to\infty}\left(1+\frac{1}{n^2}\right)^{n} = 1.$$

(3)**First way:** By binomial theorem, we know

$$\left(1+\frac{1}{n}\right)^{n^2} = 1 + n^2\frac{1}{n} + \cdots \geq n.$$

Hence

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = +\infty.$$

Second way: By the definition of e, we know there exists a $N \in \mathbb{N}$, such that for $n > N$, there is

$$\frac{e}{2} < \left(1+\frac{1}{n}\right)^{n} < e.$$

(Via choosing $\varepsilon = \frac{e}{2}$.) Hence we have

$$\left(1+\frac{1}{n}\right)^{n^2} > \left(\frac{e}{2}\right)^{n} \to +\infty \quad \text{as } n \to \infty,$$

i.e.

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = +\infty.$$

□

Problem 2.12. *Using* $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = e$ *to prove* $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e$.

Proof. By binomial theorem, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^{n} &= \sum_{k=0}^{n} C_n^k \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)
\end{aligned}$$

$$\leq \sum_{k=0}^{n} \frac{1}{k!}.$$

From the above calculation, we know for $\forall\, m > n$, there is

$$\left(1+\frac{1}{m}\right)^m \geq \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{m}\right)\cdots\left(1-\frac{k-1}{m}\right).$$

Let $m \to +\infty$, there is

$$e \geq \sum_{k=0}^{n} \frac{1}{k!} \geq \left(1+\frac{1}{n}\right)^n.$$

Hence

$$\lim_{n\to\infty} \sum_{k=0}^{n} \frac{1}{k!} = e.$$

□

Problem 2.13. *Calculate the limitation of*

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right).$$

Solution.

$$\lim_{n\to\infty} \prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right) = \lim_{n\to\infty} \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdots\frac{n-1}{n}\cdot\frac{n+1}{n} = \frac{1}{2}.$$

□

Problem 2.14. *Suppose* $\lim\limits_{n\to\infty} a_n = a$. *Prove*

$$\lim_{n\to\infty} \frac{p_1 a_n + p_2 a_{n-1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} = a,$$

where $p_k > 0$ *and* $\lim\limits_{n\to\infty} \dfrac{p_n}{p_1 + p_2 + \cdots + p_n} = 0$.

Proof. Without loss of generality, we can assume $a = 0$. (Otherwise, we can consider $a_n - a$ instead.) Since $\lim\limits_{n\to\infty} a_n = 0$, we know for $\forall\, \varepsilon > 0$, there exists $N_1 \in \mathbb{N}$, such that for all $n > N_1$, there is $|a_n| < \varepsilon/2$. Then

$$\left|\frac{p_1 a_n + \cdots + p_{n-N_1} a_{N_1+1}}{p_1 + p_2 + \cdots + p_n}\right| \leq \frac{p_1 + \cdots + p_{n-N_1}}{p_1 + p_2 + \cdots + p_n}\cdot\frac{\varepsilon}{2} < \frac{\varepsilon}{2}.$$

Since a_n is convergence, we know $\{a_n\}$ is bounded. We assume $|a_n| \leq M$ for some $M > 0$. By $\lim\limits_{n\to\infty} \dfrac{p_n}{p_1 + p_2 + \cdots + p_n} = 0$, we have

$$\left|\frac{p_{n-N_1+1} a_{N_1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n}\right| \leq \left(\frac{p_{n-N_1+1} + \cdots + p_n}{p_1 + p_2 + \cdots + p_n}\right) M$$

$$
\begin{aligned}
&< M \sum_{k=1}^{N_1} \frac{p_{n-N_1+k}}{p_1 + p_2 + \cdots + p_{n-N_1+k}} \\
&\to 0 \quad \text{as } n \to \infty.
\end{aligned}
$$

Hence there exists a $N_2 \in \mathbb{N}$ such that for all $n > N_2$, there is

$$
\left| \frac{p_{n-N_1+1} a_{N_1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} \right| < \frac{\varepsilon}{2}.
$$

Then, set $N = \max\{N_1, N_2\}$, we know for all $n > N$, there is

$$
\begin{aligned}
\left| \frac{p_1 a_n + p_2 a_{n-1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} \right| &\le \left| \frac{p_1 a_n + \cdots + p_{n-N_1} a_{N_1+1}}{p_1 + p_2 + \cdots + p_n} \right| + \left| \frac{p_{n-N_1+1} a_{N_1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} \right| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,
\end{aligned}
$$

i.e.

$$
\lim_{n\to\infty} \frac{p_1 a_n + p_2 a_{n-1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} = 0.
$$

□

Problem 2.15. *Define sequence $\{a_n\}$ by $a_{n+1} = 2a_n - a_n^2$, where a_0 is given. Discuss the convergence and divergence of $\{a_n\}$ regards the choice of a_0.*

Solution. Firstly, we have

$$
1 - a_{n+1} = 1 - 2a_n + a_n^2 = (1 - a_n)^2.
$$

By induction, we have

$$
1 - a_{n+1} = (1 - a_n)^2 = \cdots = (1 - a_0)^{2^{n+1}},
$$

i.e.

$$
a_n = 1 - (1 - a_0)^{2^n} \quad \text{for } n \ge 1.
$$

Hence,

When $|1 - a_0| < 1$, i.e. $0 < a_0 < 2$, $\{a_n\}$ is convergent, and $\lim_{n\to\infty} a_n = 1$.

When $|1 - a_0| = 1$, i.e. $a_0 = 0$ or $a_0 = 2$, $\{a_n\}$ is convergent, and $\lim_{n\to\infty} a_n = 0$.

When $|1 - a_0| > 1$, i.e. $a_0 < 0$ or $a_0 > 2$, $\{a_n\}$ is divergent, $\lim_{n\to\infty} a_n = -\infty$. □

School of Mathematical Sciences, Peking University, Beijing 100871, China.
Email address: lingwang@stu.pku.edu.cn